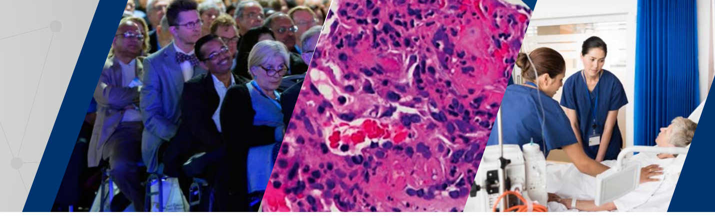

Unexplained Metabolic Alkalosis in an Athlete
Don't Sweat It

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Case Presentation

An 18-year-old woman was referred for hypokalemia and metabolic alkalosis associated with cramps and myalgias after exercise. She had recently received a college scholarship for field hockey and accelerated her exercise schedule to 3–4 hours/d in the gym and field, combining aerobic exercise and weight training. She routinely consumed 3–4 L of "Liquid IV" (electrolyte packets mixed with water) throughout and after her workouts. She denied vomiting, pica, taking diuretics, laxatives, licorice, or health supplements. Her physical examination was notable for the following: height 5′1″, weight 112 lbs., body mass index 23.5 kg/m^2, BP 92/60 mm Hg, and heart rate 58 beats per minute. She was thin but muscular. There were no oral ulcerations, and her teeth were normal. Laboratory results showed the following: sodium 135 mmol/L, potassium 2.7 mmol/L, chloride 83 mmol/L, bicarbonate 37 mmol/L, blood urea nitrogen 15 mg/dl, serum creatinine 0.9 mg/dl, glucose 81 mg/dl, and serum albumin 4.9 g/dl. The rest of the complete metabolic panel was normal. An arterial blood gas showed pH 7.47, PCO_2 50 mm Hg, and bicarbonate 36 mmol/L.

Question 1: What would be the next step in your evaluation?

A. Psychiatric evaluation.
B. Urine diuretic (and laxative if available) screen.
C. Urine Cl^- concentration.
D. Genetic testing for Gitelman.
E. Serum renin and aldosterone.

A complete discussion of metabolic alkalosis with hypokalemia is beyond the scope of this report.[1] Causes include genetic diseases, tumors, ingestions, pharmacologic agents, and surreptitious activities. With such a daunting and seemingly disparate list, it is hard to simplify the diagnostic approach. However, we present a straightforward systematic pathway to answer Question 1. We would start with the presence or absence of hypertension which when present is further categorized by levels of renin and aldosterone.[2] This patient was not hypertensive, leading us down an alternate diagnostic pathway.

Metabolic alkalosis requires two fundamental processes: Generation and Maintenance.

Generation: The serum bicarbonate concentration can increase through several mechanisms:

1. Hydrogen ion loss from the upper gastrointestinal tract, such as vomiting, nasogastric suctioning, *etc*.
2. Increased hydrogen ion loss in the urine, usually aldosterone mediated and commonly seen with diuretic use.
3. Exogenous bicarbonate or bicarbonate precursor administration, such as lactate, acetate, or citrate.
4. "Contraction alkalosis" (an often-misleading term not to be confused with volume depletion) in which the serum bicarbonate increases from fluid losses that are low in bicarbonate and rich in chloride often seen after diuresis and occasionally with lower gastrointestinal losses from a villous adenoma, infectious diarrhea rich in chloride (cholera), or laxative abuse.
5. Hypokalemia: See below.

Maintenance: As bicarbonate is filtered and normally reabsorbed, excretion of elevated bicarbonate occurs easily through: (*1*) decreased reabsorption in the proximal tubule and (2) exchanging bicarbonate for chloride in the distal tubule (see 1 below). Therefore, for metabolic alkalosis to persist ("maintained"), a process that limits kidney bicarbonate excretion is required:

1. Chloride Depletion. Pendrin is a sodium independent chloride/bicarbonate exchange protein located in the intercalated cells of the collecting duct but requires adequate chloride delivery to that nephron segment.[3] In states of extrarenal chloride depletion or effective volume depletion, proximal chloride reabsorption is augmented and thus, distal chloride delivery is decreased. This impairs

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pendrin-mediated bicarbonate excretion, and metabolic alkalosis is "Maintained."

2. A significant reduction in GFR, in which filtration alone limits bicarbonate excretion
3. A hyperaldosterone state, either:
   a. Primary or apparent hyperaldosteronism (hypertensive states not present here), or
   b. Secondary hyperaldosteronism associated with abnormal sodium chloride reabsorption as seen in Bartter and Gitelman syndrome and active diuretic use. Secondary hyperaldosteronism in effective circulating volume depletion will not lead to significant electrolyte (bicarbonate and potassium) abnormalities because proximal sodium reabsorption is augmented, decreasing distal sodium delivery and thus, limiting the substrate (sodium) for aldosterone activity. However, when distal sodium delivery is enhanced with diuretics, secondary hyperaldosteronism can generate and maintain metabolic alkalosis.
4. Hypokalemia plays a role in both generation and maintenance of metabolic alkalosis involving:
   a. Transcellular potassium/hydrogen ion shifts,
   b. Preferential aldosterone-mediated hydrogen ion (over potassium) kidney excretion, and
   c. Increased bicarbonate production through augmented ammoniagenesis.

Thus, hypokalemia can contribute to metabolic alkalosis, and metabolic alkalosis can contribute to hypokalemia.

Understanding the role that pendrin activity and chloride delivery plays in the maintenance of metabolic alkalosis, gives us the answer to Question 1, option C: Urine chloride concentration.

The urine chloride guides you to differentiate a nonhypertensive metabolic alkalosis to either:

1. High (>20 mmol/L) associated with tubulopathies (Bartter and Gitelman) and active diuretic use.
2. Low (<20 mmol/L) associated with extrarenal chloride depletion or distal delivery.

The patient's urine chloride was <10 mmol/L indicating 2. Her urine diuretic and laxative screen were negative. In addition, her metabolic abnormalities resolved between workouts. This brings us to:

Question 2: What would be the next step in your evaluation?

A. Psychiatric evaluation.
B. Serum renin and aldosterone.
C. Genetic evaluation.
D. Repeat urine diuretic and laxative screen.
E. Phone a friend.

This patient has a "low chloride" metabolic alkalosis in which the generation and maintenance were a mystery. Diuretics, laxatives, and vomiting (bulimia) can each generate and maintain a metabolic alkalosis. In vomiting and chloride-rich laxative abuse, the urine chloride would be expected to be low, but with recent use of diuretics, it would be high. The patient denied these activities, but as they would be surreptitious, this would be expected. These behaviors usually accompany a body dysmorphic disorder, not apparent in her. Urine diuretic and laxative screening were negative, but a clever patient may time their use with blood and urine sampling. Without constant observation, these diagnoses are almost impossible to confidently rule out and this psychiatric evaluation and repeat urine screening (options A and D) seem reasonable. Serum renin and aldosterone (option B) will not likely help in the diagnosis in this case as they will be elevated in both tubulopathies and effective volume depletion.

So, what generates her metabolic alkalosis with exercise? She consumes several liters of "Liquid IV" during workouts. We contacted the company to determine if it contained any source of base but received: "We are unable to share exact values since this would be proprietary information." We sent a specimen of Liquid IV to Litholink (Itasca, IL), a company that specializes in evaluating urine stone risk analysis), which determined that there was approximately 50 mmol net alkali/liter: Generation Solved.

However, "maintenance" remained a mystery. She did not have any obvious kidney or gastrointestinal chloride losses, what was the source of her chloride deficiency ("Phone a friend" option E)? She sweats extensively during her workouts. While sweat contains sodium chloride, the amount is regulated by aldosterone through epithelial sodium channels, similar to the kidney (Figure 1B). We then obtained a sweat chloride twice, which was approximately 40 mmol/L. Patients with cystic fibrosis (CF) have a sweat chloride >60 mmol/L with borderline values 40–60 mmol/L and only if <30 mmol/L is CF unlikely.[4] No other clinical parameters suggested this diagnosis, but we sent her blood for CF genetic testing (correct option C) and determined she was a heterozygote for the CF transmembrane conductance regulator (CFTR) gene c.2991 G>C p. (L997F). CF is an autosomal recessive disease. Carriers (heterozygotes) of autosomal recessive conditions do not typically manifest the phenotype that is characteristic of homozygotes. However, there are over 50 different signs and symptoms in 10 different organ systems in CF carriers that while not at the same prevalence or severity as in CF homozygotes are not normal compared with individuals without a CF mutation.[5] A recent cohort of 29,907 CF carriers and 299,070 controls found an increase in fluid and electrolyte disorders in the CF carriers, including volume depletion, hypokalemia, and metabolic alkalosis.[6] A case similar to ours was reported in a CF homozygote, a chef and who developed severe hypokalemic metabolic alkalosis, but only in the summer when sweating excessively in a hot kitchen.[7] There are other case reports of exercise-induced hypokalemic metabolic alkalosis, but CF has not been suggested.[8,9] One (asymptomatic) was identified through an exercise research project. The other was a patient who had a seizure after a workout. The authors mentioned CF (but not trait) in the differential diagnosis and the possibility of high sweat chloride loss later but was not measured. In both cases, the patients were vegetarians,

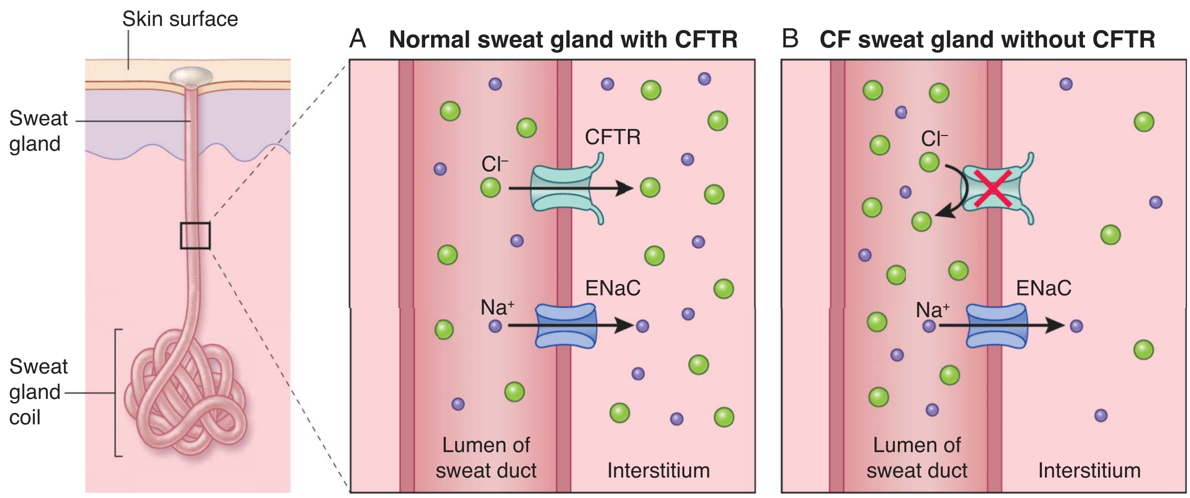

Figure 1. Abnormal chloride transport in sweat glands of patients with cystic fibrosis. Defective sweat chloride (Cl) transport in patients with CF: Sodium (Na) and Cl are normally reabsorbed from the sweat duct into the interstitium: Na through ENaC channel channels and Cl through a channel that requires CFTR as demonstrated in (A). CFTR is lacking in patients with CF (B) leading to excessive sweat Cl. This provides a sweat gland lumen negativity that limits Na transport (reabsorption) as well. These effects lead to excessive sweat Cl loss, which can affect the patient's volume and acid base status. CF, cystic fibrosis; CFTR, CF transmembrane conductance regulator; ENaC, epithelial sodium.

which is a diet high in net alkali. This is similar to our patient who ingested an oral replacement fluid high in alkali. We would posit that these previously reported unusual cases may also represent a carrier state for CF and should alert providers to pursue genetic testing.

The CF gene encodes for CFTR, a protein that allows chloride transport (in conjunction with epithelial sodium mediated sodium transport) from the sweat gland ductal lumen into the interstitium[10] (Figure 1). Interestingly, in some patients with CF, renal pendrin is downregulated (also from a relative deficiency in CFTR), leading to excessive urinary chloride loss, cleverly termed "renal tubular alkalosis."[11] With a low urine chloride, we felt her high sweat chloride alone explained her chloride depletion, leading to genetic testing: Diagnosis and Maintenance solved.

Case Summary

We believe that this patient's metabolic alkalosis was generated by base intake from Liquid IV along with "contraction" from large volumes of sweat rich in chloride and low in bicarbonate. It was maintained by heterozygote CF–induced sweat chloride loss, leaving her total body chloride deficient, thus rendering pendrin ineffective. She decreased her training routine, replaced Liquid IV with potassium chloride, and her metabolic alkalosis has not recurred.

Disclosures

Disclosure forms, as provided by each author, are available with the online version of the article at http://links.lww.com/CJN/B908.

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Author Contributions

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